

Unit 2 : Customer-Centric Product Development

Mission statement and technical questioning, Technology forecasting and S curve, Customer needs and satisfaction, Customer needs - types and models, Tools for gathering customer needs, Customer population and market segmentation, Customer involvement in product development.

Mission statement and technical questioning

The result of technical questioning procedure should be clear statement of design teams mission. The mission statement sometime called as a market attack plan, a vision statement or product plan. The mission statement typically first written by a project manager or team and distributed to all development team members to explain what they will be doing. The mission statement also has another use, when interviewing customers meeting with potential suppliers or carryout design reviews, the mission statement can serve as the calling cards to explain, what the end goal.

Mission statement	## Product
Product Description	One concised and focused sentence
Key business	Schedule
Goals	Margin/Profit, Advancement of human needs
Assumptions	Key assumption or uncontrolled factors to be confirmed by customers
Avenues for creative design	Identifies key areas for innovation

Mission statement template used to write any example

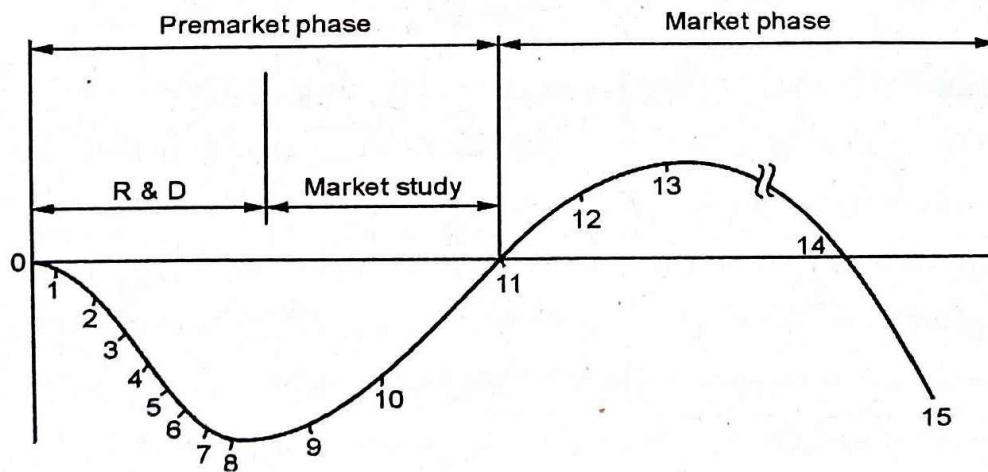
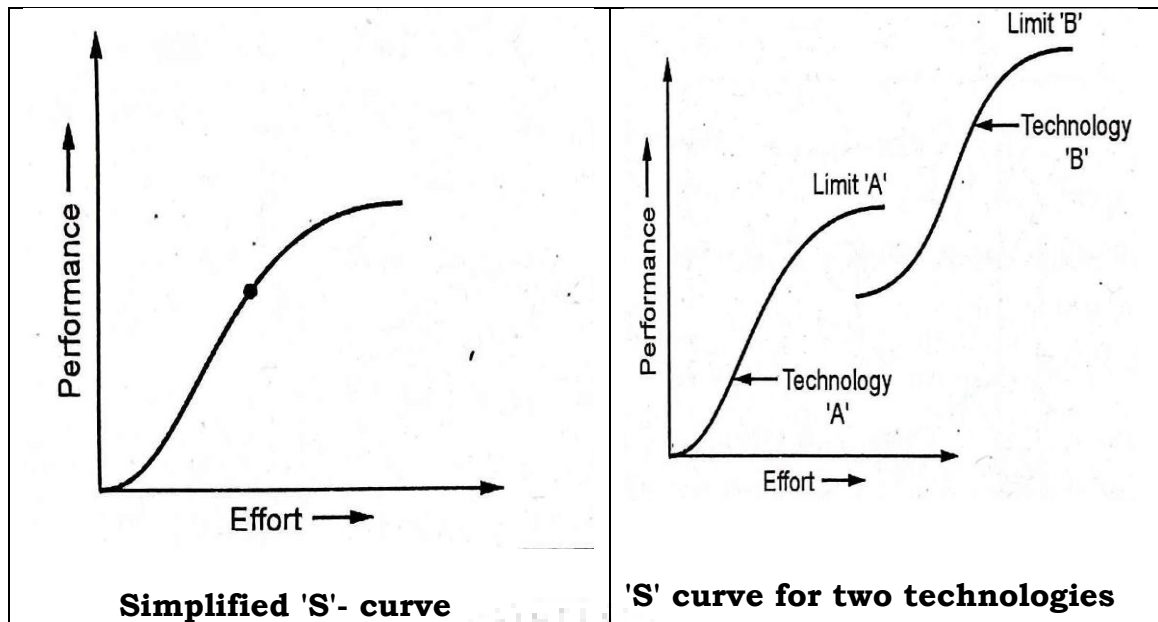
Technical Questioning

- The first step in task classification is usually to gather additional information. The questions like following need to be asked and answered, not once but continuously through life cycle of design process.
- What is the problem really about?
- What expectations and desired are involved?
- Are the stated customer needs, functional requirement and constraints truly appropriate ?
- What avenues are opened for creative design and inventive problem solving?
- What Characterstics/Properties must the product have?
- What Characterstics/Properties must the product not have ?
- What are the technical and technological conflicts inherent in the design task ?

- These questions force the design team to think critically, the first restate a design task in more precise way for the project or subset currently under consideration. Immediately answering the questions is not the focus.

Technology Forecasting and S-Curve

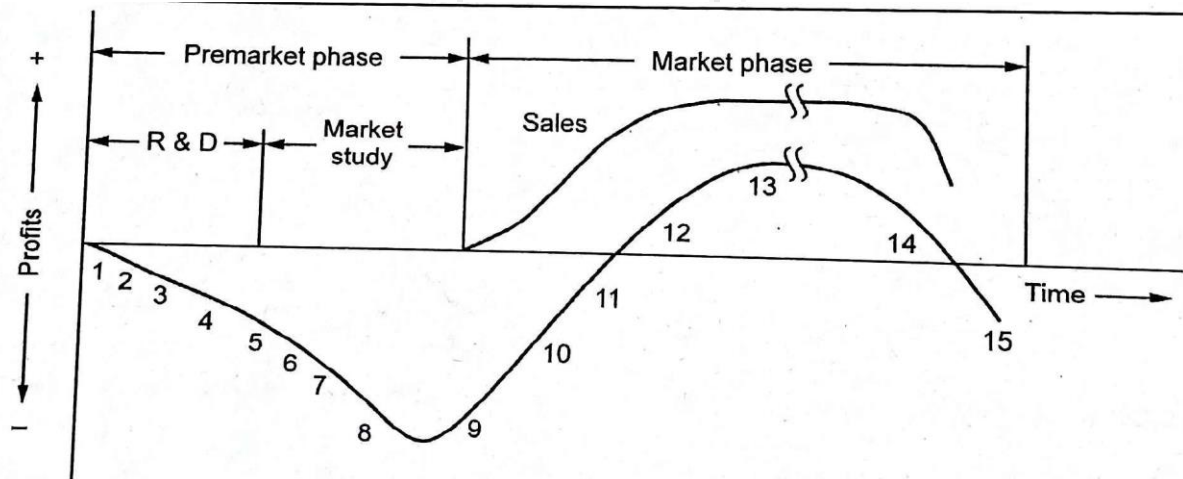
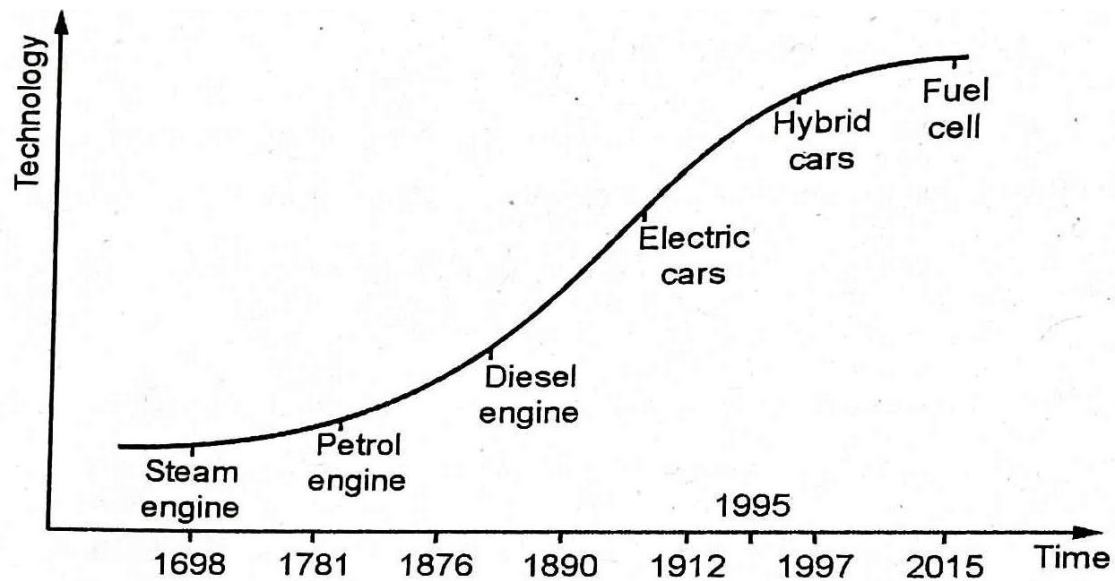
- S-curves are commonly used in technology forecasting. They show the path of product performance in relation to time or investment in research and development.
- The development of new technology follows 'S' shaped growth curve. In it's early stage progress in technology tends to be limited by the lack of ideas.
- A single good idea can make several other good ideas possible and the rate of progress becomes exponential as indicated by steep rise in the performance that creates the lower steeply rising curve of the 'S'.
- During this period, a single individual or small groups of individual can have pronounced effect on the direction of technology.
- Gradually the growth becomes more nearly linear when fundamental ideas are in place and technical progress is concerned with filling gaps between key ideas.
- This is period, when commercial exploitation of flourishes, specific design market applications and manufacturing occur in a field that has not yet settled down.
- However, with time technology begins to run dry and improvement come with greater difficulty.
- Now the market become stabilized, manufacturing method become fixed and more capital is expended to reduce cost of manufacturing.
- The business become capital intensive. The emphasis is on production known how and financial expertise rather than scientific and technological expertise.
- The maturing technology grows slowly and it approaches a limit.
- The limit may be set by social considerations such as the fact that the legal speed of automobiles is set by safety and fuel economy and it may be true technological limit.
- The success of technology based company lies in recognizing when the core technology on which the company's products are based is beginning to mature and through an active R and D program.
- To do this, company must manage across a technological discontinuity and new technology must replace the existing one.
- Technology usually begin to mature before profits top out, so there is often management reluctance to switch to new technology with its associated costs and risk, when business is doing well.
- Company are always lookout for the possibility for technology insertion because it can give them a big advantage over the competition.



Expanded view of product development cycle

- 1.Idea Generation 2.Idea Evaluation 3.Feasibility analysis 4.Technical R & D
- 5.Product (Market R & D) 6.Preliminary Production 7.Market Testing
- 8.Commercial Production 9.Product introduction 10. Market Development
- 11.Rapid Growth 12.Competative market 13.Maturity 14.Debate

E.g.

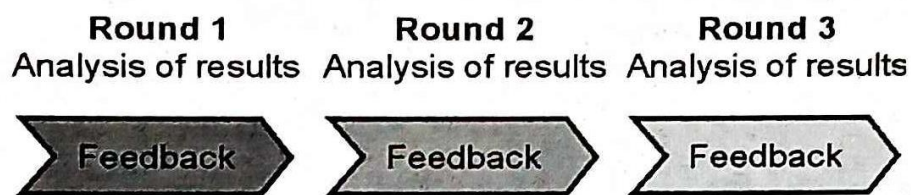


- In 1698, English engineer Thomas Savery patented steam pump. Thomas Newcomen has worked to develop first commercial true steam engine in 1712. In 1781, Scottish engineer James Watt patented a steam engine that produced continuous rotary motion.
- Then 1876 in Germany first practical otto engine (petrol engine) had been developed by August otto. Because of lower efficiency of steam engine i.e. 10% to 15%, the gasoline engines has taken the market and replaced the steam engine. The gasoline engines have efficiency as high as 35 to 40%.
- Also I.C. Engine has low weight to power ratio as because of its compact design. Then because of its low efficiency and low load carrying capacity, for heavy duty applications Rudolph Dierce have introduced diesel engine in 1890, which has very high expansion ratio that gasoline powered I.C. engine.

- Now because of increased in pollution and government rules and regulations, electronic vehicles has been commercially developed in the year 1995 and then in order to maximize the performance of electric cars have been combined with fuel powered engine and hybrid vehicles had been introduced in next few years.
- At the end of 2015, fuel cell vehicles have been launched by major automobile companies. Fuel cell critically ensures least minimum emission levels.

Delphi Forecasting Method

- The Delphi method is a forecasting process framework based on the results of several rounds of questionnaires sent to a panel of experts.
- Several rounds of questionnaires are sent out, and the anonymous responses are aggregated and shared with the group after each round.
- The experts are allowed to adjust their answers in subsequent rounds. Since multiple rounds of questions are asked and the panel is told what the group thinks as a whole, the Delphi method seeks to reach the correct response through consensus.



The Delphi method generally involves the following stages:

- A panel of experts is assembled.
- Forecasting tasks/challenges are set and distributed to the experts.
- Experts return initial forecasts and justifications. These are compiled and summarized in order to provide feedback.
- Feedback is provided to the experts, who now review their forecasts in light of the feedback. This step may be iterated until a satisfactory level of consensus is reached.
- Final forecasts are constructed by aggregating the experts' forecasts.
- At the end of each comment session, all questionnaires are returned to the facilitator who decides if another round is necessary or if the results are ready for publishing. The questionnaire rounds can be repeated as many times as necessary to achieve a general sense of consensus.

Customer Needs-Types and Models

Types of customer needs : There are different types of customer needs, such as:

- a) Direct needs b) Latent needs c) Constant needs d) Variable needs
- e) General needs f) Niche needs

a) Direct needs: These are the needs that when asked about product, customer's have no trouble declaring as something they are concerned about. e.g. Calculations of engineering problems is the direct need of engineering students, which intend then to purchase scientific calculators.

b) Latent needs : These are the needs that are typically not directly expressed by the customer without probing to find the latent need. Latent needs are better characterized as customer needs, not of product but the system within which the product operates other products, services or actions currently satisfy the needs directly. Yet these needs might be fulfilled with a developing product and doing so can provide competitive advantage. E.g. - Convenience, productivity, usability, experience, efficiency and quality of life are the example of latent needs. Convenience: Products and services that save the customer time and efforts. In the 1960s, few customers have asked for a faster oven because they would have assumed that it would burn the food.

c) Constant needs: These need are intrinsic to the task of product and always will be when a product is used, this need will always be there. Such needs are effective to examine with customer needs analysis. E.g.: In every manufacturing organization, various machines and equipment requires lubricating oil and grease for reducing wear and tear of machine parts.

d) Variable needs: These needs are not necessarily constant, if a technological change can happen, these needs go away. E.g. - Digital photography eliminates customer needs of long film storage life.

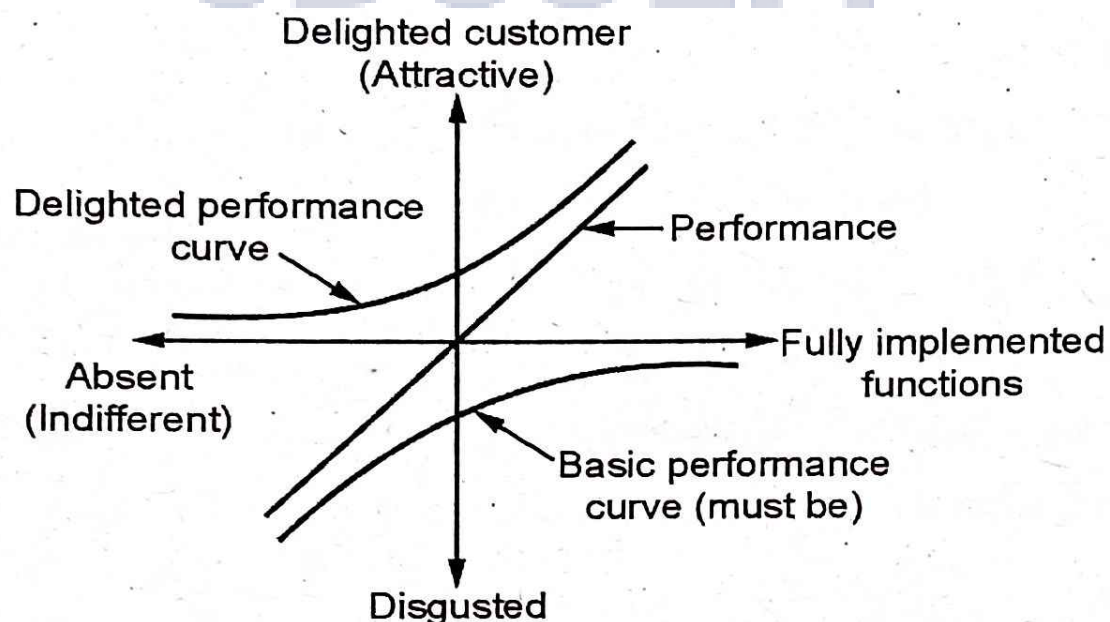
e) General needs: These needs apply to every person in customer population. E.g. - Every customer purchasing new vehicle, want good performance characteristics with minimum cost.

f) Niche needs: These needs apply only to smaller market segments within the entire buying population. E.g. The general tooth paste available in the market is required on mass by all but there is a niche market of customers, which is segment of main market that demands specialized tooth paste sensitive to their teeth.

Customer Satisfaction or Voice of the Customer

- It is difficult to understand to understand, what customer wants, where several issues confound what designer needs to know.
- For example, a customer must understand the product that is being desired.

- Typically, this understanding is being conveyed by showing the current product model to a customer and asking for preference information about its features.
- The product model shown to the customer is not same as the new model to be designed.
- This is a particularly difficult problem for products with rapidly changing technology or for new product domains where technologies are not well established.
- Another problem is that customers, only discuss the failure of the products, what they do not like about their experience in using the product.
- This difficulty is graphically explained by considering customer satisfaction versus any product function known as the Kano diagram.
- The Kano diagram (Shiba, Graham and Nalden 1993) depicts the idea that one might consider customer satisfaction on a scale from disgusted to delighted.
- Similarly consider product function from being absent to being fully implemented.
- If the product function is good for customer need being considered, one could plot a 45° line, which would indicate the nominal supplied satisfaction for any satisfaction level of the function.
- This is known as linear quality. On other hand one could also plot a lower curve that would indicate the minimum basic level of satisfaction that a customer presumes must exist for the function implementation level. This plot is known as basic performance.
- Similarly can plot an upper curve that would indicate delighted state that a customer is satisfied.



Kano diagram of customer satisfaction

Customer Need Models

- In this analysis, a statistical representation of the product, customer needs is the desired result.
- This results can be as simple as the list of needs from interviews from the customers. The resulting needs might be increased with importance weighting determined through questionnaire with the importance of each need determined independently of the others as statistical distribution and an average weight.
- At more advanced level, the customer needs can be uncovered as list of needs, each need complete with a range of possible target values, any value of which a design team might aim to meet.
- This representation may be considered a vector space, where the positions of all members in the market population can be represented within this vector space as a probability distribution.
- Every person in the market has an ideal set of target values on the list of needs and every person is different.
- This probability space might be examined for large regions of high probability (Cluster analysis) which if the each product is targeted in a centre of such cluster the development effort is offered some assurance of larger success.
- Often customer needs list of a coupled dozen need is sufficient for development purpose. This guidelines is often true for studies, such as understanding market position.
- On the other hand for detailed embodiment, questions, design team often would like to know what the customer preference is for different selection choices.

Tool for Gathering Customer Needs

Interviews

- Here a design team members discusses the need with single customer, one at a time such interviews are held in the customer environment, where the customer uses the product.
- Design team members record the response. These records can be supported with video or audio tapes.

Questionnaires

In this tool, team develops a list of criteria it thinks is relevant to the customer's concerns. It then ranks product on this criterion.

Focussed Groups

- In this tool, a moderator facilitates a session with group of customers. Usually, this session is held in product developer's environment, typically inside a room with a two way mirror, so that the design team can observe the customer during a session.
- Effective focussed group practice is to have the moderator asked questions such as "why" to uncover the need of customers. Typically

several sequential focus groups sessions are held with customer need analysis of the customer's statement constructed between each session.

- This analysis provides two essential results.
- An understanding of what to probe (for the moderator) and development of need list (for the design team).
- Be the customer Design team will also travel to the location where their competitor's product is used and acts as a customer.
- This activity is typically executed by the design team, working hand in hand with customer's needs.

Conducting Interviews

- Like/ Dislike Method: The simplest approach that proves reasonably effective in practise is to interview customers as they use the product.
- During the interviews a team member asked the customers to describe, what they like or what not like about the product.
- During the process more detailed questions may be asked to explore different facets of the product. At the same time, a reasonable list of customer needs is developed. While the customers describes and actively use the product, the interview should ensure that they describe not only what they do not like but also they do like.
- Asking about the likes ensures design team understand what is expected in the product and what must be provided.
- Asking about the dislikes in the product establishes of what the design team can provide to delight the customers as it currently not being expected.
- Some general hints for an effective customer interaction (Unrich and Eppinger, 1995) are:

Go with the flow

- Wherever the customer takes you, follow along and ask why and how questions.

Use visual stimuli and props

- Bring models of new concepts, competitors product.

Have the customer demonstrate

- Don't just ask about the product, human language is so expressive, seeing the need in action will permit much better understanding.

Customer Data : Project/Product Name			
Customer Address		Interviewers	
Type of user		Date	
Question	Customer statement	Interpreted need	Importance
Typical uses			
Likes			
Dislikes			
Suggested improvement			

Like/ Dislike method data collection form

Grouping interpreted need

- Affinity Diagram Method: Having formed interpreted needs list, for sufficient customers, the information must be compiled into a list of interpreted customer needs to which a product will be re-designed, this is done with affinity diagram method.
- To group the needs, each interpreted need with associated customer identification is copied onto a post-it note.
- Then one takes the post-it notes and affixes the first one on white large board.
- The next customer need is compared with the first, if it is different, a new column is constructed on the board.
- If the same then post-it note is placed under similar statement. The same process is repeated for all post-it notes.
- Thus, method will create a grouping of customers needs that reflects customer's desires, demands and importance.
- This is generally not the best as it is expressed in original words of the customers and the reasoning behind why the customer made the statement can be lost.
- On the other hand method is easy to implement.

Customer Population and Market Segmentation

- Customers can be successfully thought of as belonging to distinct market segment.
- Dividing a market into segment allows the firm to consider the actions of competitors and strength of firms.
- There are four types of market segmentation:

1) Demographic segmentation: This is one of the simplest and most widest type of market segmentation. Most of the companies use it to get the write population of customer in using their product. Demographic segmentation has variables such as age, gender, income, family size, religion, nationality. Demographic segmentation can be seen applying in automobile market. E.g.

- Maruti has low price bracket and therefore manufactures people's driven car. Audi/BMW have the highest price bracket, so it target high end buyers.

2) Behavioural segmentation: This segmentation is based on population's behaviour, their usage and decision making patterns. E.g. Blackberry was launched for users who were business people. Samsung was launched for users who like android and like various applications for free price, whereas Apple launched for premium customers who want to be a part of unique and popular niche.

3) Psychographic segmentation: Is the segmentation based on life style of the people, their interest, activities. This is quite similar to behavioural segmentation. E.g. - Arrow cloth market itself to the premium office lifestyle, where probably bosses or super bosses shop for the sharp clothing.

4) Geographic segmentation: This divides people on the basis of geography. Customers will have different needs based on geography, they are located in. E.g.:

1) In cold countries, the same company might be marketing for heaters whereas in hot countries. The same company might be targeting air-conditioners.

2) RO water purifiers used for non-municipal area and UV water purifier for municipal area.

Product Development Economic Analysis

- When completing an analysis of the market viability of development project, initially a project manager must assess the cost of development versus the expected future revenues. Expected future revenues must compensate for the development investment made today. Two fundamentals consideration therefore are apparent..Risk and the Time Value of Money.
- A development project economic analysis consist of estimating and establishing values for economic variables and then evaluating the expected economic profit, performance of the new development project. Typical economic performance criteria includes:
 - 1) Break Even Point (BEP): The time until the investment cost in recovered and the project start making money (No Profit No Loss).
 - 2) Return on Investment: Over the expected sales life in the market place, the amount of profit gain compared to the initial investment.
 - 3) Investment Risk: The probability that the market analysis is not correct and there will be less return on investment.
- To conduct such analysis yearly Net Present Value (NPV) cash flow model can be used. That is one can establish the development cost in today's currency value. Then a time line is projected in to future years, over the life of product in the market place.

Questions

1. Discuss mission statement and technical questioning in context to product.
2. Explain 'S' curve with suitable example.
3. Explain various customer needs with feedback.
4. Discuss customer satisfaction.
5. Explain tools for gathering customer needs.
6. Enlist and explain types of market segmentation.
7. Discuss how mission statement and technical questioning helps to design better. Elaborate in detail.
8. What is the problem really about?
9. What expectations and desires are involved?
10. Are the stated customer needs, functional requirements and constraints truly appropriate.
11. What avenues are open for creative design and innovative problem solving?
12. What characteristics, properties must the product have?
13. What characteristics, properties must the product not have?
14. What are the technical and technological conflicts inherent in the design task.

